

The Origins of Modern Computing

CSCI 264 • Lecture 11 • Dive into Systems §5.1

Monday, September 21, 2026

Listen for three threads

- **Women in early computing** — who did the work, and who got the credit
- **The military** — who paid for the machines, and why
- **John von Neumann** — and the document that put his name on an architecture

A timeline

1837	Babbage & Lovelace: the Analytical Engine
1937	Shannon's thesis; Turing's machine
1943	Colossus at Bletchley Park
1944	Harvard Mark I
1945	ENIAC completed; the <i>First Draft</i> circulated
1946	ENIAC unveiled; Turing's ACE report

“Computer” was a job title

- 1700s–early 1900s: a *computer* was *one who computes*
- A person doing calculations by hand, usually one of a large team
- Most often a woman
- Every machine today was built to replace a room full of these people

Babbage and Lovelace



Ada Lovelace, watercolour by A. E. Chalon.

Wikimedia Commons.

- The **Analytical Engine**: a mechanical, general-purpose design
- Ada Lovelace published the first algorithm meant for a machine
- “First computer” is *retrospective*
- Later designers mostly didn’t know their work

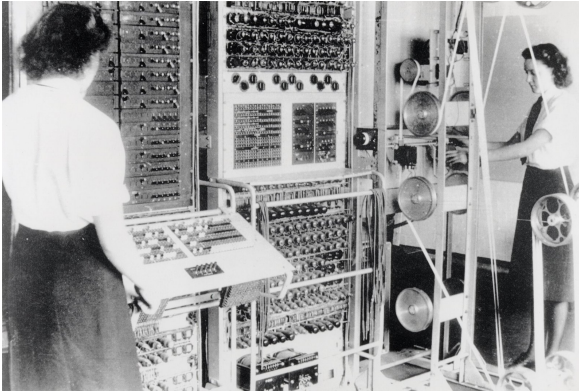
A primordial soup of ideas



Alan Turing. Wikimedia Commons.

- **1937, Claude Shannon:** Boolean logic can be built from switches
 - the basis of all digital circuits
- **1937, Alan Turing:** the Turing machine
 - what a computer can and cannot do
- Neither of them built a machine

Colossus and the Wrens



Colossus at Bletchley Park, 1943. Wikimedia Commons.

- 1943, Tommy Flowers, Bletchley Park
- Arguably the first programmable, digital, fully electronic computer
- Operated by the **Wrens**
- Noted for cryptographic ability ...
- ... never made cryptographers

Meanwhile, in the United States

- **Harvard Mark I** — Howard Aiken, a U.S. Navy Reserve commander
- **ENIAC** — Eckert and Mauchly, University of Pennsylvania, 1945
 - Digital, fully electronic, programmable, general purpose
 - Decimal, not binary
- Built for **U.S. Army Ordnance**
- Purpose: **artillery firing tables**

The people ENIAC was built to replace

- $\approx$ 200 women computing trajectories by hand at Penn
- Slide rules, desk calculators, a differential analyzer
- One trajectory: 20 minutes to several days
- Kay McNulty's appointment: **“subprofessional”** grade

J. S. Light, "When Computers Were Women," *Technology and Culture* 40:3 (1999).

The first programmers

- The human computers became the first programmers
- Because programming was considered **clerical**
- The machine was the achievement; instructing it was typing

Grace Hopper



Cropper, Krishnan, Hopper, Rothberg at a UNIVAC I console,

1957. Computer History Museum 102741216.

- Programmer on the Mark I; wrote its manual
- First compiler: **A-0**
- **FLOW-MATIC** → COBOL
- A commissioned officer in the **U.S. Naval Reserve**

The “first bug”

9/9

0800 Ancon started
1000 stopped - ancon ✓

1300 (032) MP-MC
(033) PRO -

conv 2.130476415
2.130676415

Relays 6-2 in 033 failed spiral speed test
in relay 11.000 test.

Relays changed
Started Cosine Tape (Sine check)
1525 Started Multy Adder Test.

1545

Relay #70 Panel F
(moth) in relay.

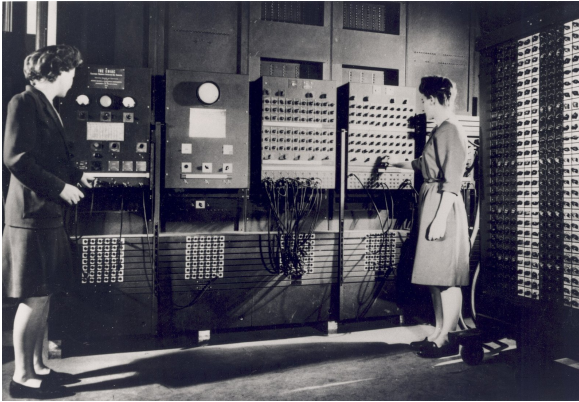
First actual case of bug being found.
1630 Ancon started.
1700 closed down.

Relay 3145
Relay 3370

- Harvard Mark II log, 1947
- A moth, taped in
- Not the first use of “bug”
- Probably not written by Hopper

Harvard Mark II logbook, 1947. Wikimedia Commons.

The ENIAC Six



ENIAC at the Moore School. U.S. Army photo, Wikimedia Commons.

Jean Jennings Bartik
Betty Snyder Holberton
Kay McNulty
Frances Bilas Spence
Marlyn Wescoff Meltzer
Ruth Lichterman Teitelbaum

- Given wiring diagrams, not a manual
- Invented flow charts, subroutines, nesting

The credit did not follow

- February 15, 1946: ENIAC unveiled, running Jennings and Snyder's program
- Congratulations went to Eckert and Mauchly
- *New York Times*: the work of "100 trained men"
- Photo captions: "setting switches," "plugging cables"
- Reused as an Army recruiting ad — with the women cropped out

ENIAC's first real job

Not a firing table.

A feasibility calculation for a **thermonuclear weapon**.

John von Neumann



Von Neumann's grave, Princeton Cemetery.

Photo: M. West.

- Budapest, 1903; Institute for Advanced Study, 1933
- Mentor to Turing at Princeton
- Manhattan Project → Mark I, ENIAC
- ENIAC's first run was his calculation

The First Draft of a Report on the EDVAC

- June 1945
- EDVAC: ENIAC's successor, proposed by Eckert and Mauchly
- **Binary**, not decimal
- **Stored program**: instructions and data both live in memory

Instructions are just more bits

One name on the cover

- A draft, unfinished — blanks where the references should go
- Months of group work at the Moore School
- Circulated by **Herman Goldstine, the Army's liaison**
- Eckert and Mauchly furious: circulation undermined their patent claims
- Turing's ACE report, 1946 — but von Neumann's came first

Afterword

- **1952** — Alan Turing convicted of “gross indecency”
Clearance revoked; forced hormone treatment; dead in 1954
- **1997** — the six ENIAC programmers inducted into the WITI Hall of Fame
Fifty-one years later; Ruth Lichterman Teitelbaum had died in 1986

Two minutes, on your own

Of everything today,
what surprised you most — and why?

You won't hand it in.

Discussion

Programming was given to women because it was the “clerical” half of the work. Today it is the high-status half.

What changed? What does that tell us about how a field decides which work is valuable?

Discussion

Almost every machine today was built for a war, with an army's money, to solve an army's problem.

Do we get this architecture without WWII — or just later, and differently?

Discussion

Eckert and Mauchly built it. Six women programmed it.
Von Neumann's name is on it.

**Is “von Neumann architecture” the wrong name?
Does it matter?**

Discussion

The Wrens were recorded as having real cryptographic ability, and stayed operators. Nobody wrote down a decision to exclude them.

So what would have had to be different?

Further reading

- Jennifer S. Light, “When Computers Were Women,” *Technology and Culture* 40:3 (1999), 455–483
- Janet Abbate, *Recoding Gender*
- Kathy Kleiman, *Proving Ground* and the documentary *The Computers*
- LeAnn Erickson, *Top Secret Rosies* (PBS)

Wednesday: we open the box — the five units of the von Neumann architecture.